

Agri Lesson 13: Food Quality, Safety, and Standards

Comprehensive Study Notes translated and structured from Sinhala educational resources

1. Introduction to Food and Quality

What is Food? (ආහාරයක් යනු කුමක්ද?)

Food is any product of plant or animal origin containing one or more nutrients that provides energy for the body's growth and metabolic processes, while supporting the immune system.

Why Do We Consume Food?

Humans consume food primarily to fulfill daily nutritional requirements and maintain proper physiological health.

Key Characteristics of Consumable Food

- Organoleptic (sensory) characteristics such as color, aroma, taste, and texture must be at a level acceptable to the consumer.
- Must be completely free from harmful chemical substances or other contaminants.
- Consumption must not lead to diseases or other infections.

Food Quality (ආහාරයක තත්ත්වය)

Food quality refers to the condition where characteristics inherent to a food item are at a level acceptable to the consumer. Inherent characteristics include color, taste, aroma, texture, and nutrient content. The complete absence of harmful chemicals (e.g., pesticides) and foreign matter (e.g., animal feces, stones, sand) is vital.

However, physical characteristics (aroma, texture, color, external appearance) and chemical characteristics (nutrients) can change due to various factors:

- Moisture (තෙතමනය) & Temperature (උෂ්ණත්වය)
- Mechanical damage (යාන්ත්‍රික හානි) & Time (කාලය)
- Enzymatic activity (එන්සයිමය ක්‍රියාකාරීත්වය)
- Rancidity (මුඩු වීම) & Fermentation (පැසීම)
- Microorganisms (ක්ෂුද්‍ර ජීවීන්) & Macro-organisms (මහා ජීවීන්)

2. Importance of Standards and Regulations

To ensure high quality and safety in food supplied to the public, governments establish food control systems. Standards represent a collection of rules established to protect consumers.

Key Benefits of Quality Standards

- **Consumer Protection:** Prevents purchase of sub-standard goods, protects against fraudulent trade practices, and avoids misinformation.
- **Producer Benefits:** High demand for quality food creates business advantages, builds identity, excellence, and consumer trust.
- **Public Health:** Consuming safe, high-quality food builds a healthy population, reducing medical expenditures.
- **Efficiency & Waste Reduction:** Increases resource utilization efficiency and minimizes spoilage of perishable agro-products (e.g., milk).
- **Economic Growth:** Increases export potential of high-quality food products, earning valuable foreign exchange.

3. Food-Related Acts and Regulatory Bodies in Sri Lanka

Key Legislative Acts

- **Food Act No. 26 of 1980 (Amended in 1991):** The primary legislation governing food laws in Sri Lanka, operated under the Ministry of Health and Nutrition.
- **Consumer Affairs Authority Act No. 9 of 2003:** Replaced the earlier Consumer Protection Act No. 1 of 1979 to ensure consumer protection and market regulation.

Regulatory Institutions & Standards Framework

Institutions (ආයතන)	Standards Categories (ප්‍රමිති)
• Sri Lanka Standards Institute (SLSI) • Ministry of Health and Nutrition • Consumer Affairs Authority (CAA) • Internal Trade Commissioner Department	• Sri Lanka Standards (SLS) • Company Standards • Regional Standards • International Standards • Corporate/Institutional Standards

4. Certification Systems and International Standards

SLS (Sri Lanka Standards) Certification

A quality assurance certificate issued by the SLSI indicating that a product is manufactured according to established standards. Formulated by SLSI technical committees, the SLS mark builds consumer confidence and market demand. (Example: *SLS 183* for bottled carbonated beverages).

Mandatory Requirements for SLS Certification:

1. Production or service must strictly conform to the relevant standard.
2. A quality assurance management system must be operational within the organization.
3. Agreement to pay 0.05% of total annual revenue to the SLSI.

ISO (International Organization for Standardization)

ISO 22000 is a globally recognized food safety management standard incorporating CODEX regulations. It ensures complete traceability across the supply chain, strong inter-communication among stakeholders, systematic management, and reliable prerequisite programs (GMP, GHP).

5. Management Systems for Food Quality Assurance

Ensuring food safety requires structured managerial systems implemented across agricultural production and manufacturing:

- **Good Agricultural Practices (GAP):** Controls physical and chemical changes from farm to factory. Focuses on site selection, planting material, integrated pest and nutrient management, water quality (pH, salinity), farm hygiene, and post-harvest technology.
- **Good Manufacturing Practices (GMP):** Ensures factory premises meet sanitary requirements, non-polluting locations, proper layout, and easy-to-clean stainless steel machinery.
- **Hazard Analysis Critical Control Point (HACCP):** A systematic preventive approach to food safety addressing physical, chemical, and biological hazards.

The 7 Principles of HACCP

1. Conduct hazard analysis (උපද්‍රව / අනතුරු විශ්ලේෂණය කිරීම)
2. Determine Critical Control Points (CCPs) (අවධි පාලන ලක්ෂ්‍ය නිර්ණය කිරීම)
3. Establish critical limits (අවධි සීමා ස්ථාපිත කිරීම)
4. Establish monitoring procedures (අවධි පාලන ලක්ෂ්‍ය නිරීක්ෂණය කිරීම)
5. Establish corrective actions (නිවැරදි කිරීමේ උපක්‍රම ස්ථාපිත කිරීම)
6. Establish verification procedures (නිවැරදි කිරීමේ ක්‍රියාවලිය ස්ථාපිත කිරීම)
7. Establish record-keeping and documentation (නිවැරදි වාර්තා තබා ගැනීම)